

Derivation of the Energy-Momentum Tensor from the Hilbert Action

Ben Thomas · February 28, 2026

Context

The conceptual structure of this derivation follows material introduced in advanced coursework at the Universitat de Barcelona. All detailed calculations, formal development, and typesetting are the independent work of the author.

1. Electrodynamic Energy-Momentum Tensor

The energy-momentum tensor (TEM) of Electrodynamics can be obtained from the Maxwell action using the functional or variational definition provided in General Relativity. In our metric (and in Heaviside-Lorentz units), we must find:

$$T^{\mu\nu} = F^\mu{}_\lambda F^{\nu\lambda} - \frac{1}{4}g^{\mu\nu}F^{\alpha\beta}F_{\alpha\beta} = -F^\mu{}_\lambda F^{\lambda\nu} - \frac{1}{4}g^{\mu\nu}F^{\alpha\beta}F_{\alpha\beta}.$$

We verify that with the present metric signature, the result is opposite in sign to the one given in standard Electrodynamics. Furthermore, we verify it is symmetric: $T^{\mu\nu} = T^{\nu\mu}$.

Given the definition of the energy-momentum tensor as the variation of the action with respect to the metric:

$$\delta S_M = \frac{1}{2} \int d^4x \sqrt{-g} T^{\mu\nu} \delta g_{\mu\nu}$$

we can equate this quantity δS_M to the variation of the Maxwell action, taking into account that in General Relativity we transform the invariant $d^4x \rightarrow d^4x \sqrt{-g}$:

$$\begin{aligned} \delta S_M &= \frac{1}{2} \int d^4x \sqrt{-g} T^{\mu\nu} \delta g_{\mu\nu} = \delta \int d^4x \sqrt{-g} \mathcal{L}_M = \int d^4x \delta (\sqrt{-g} \mathcal{L}_M) \\ &= \int d^4x \left[\delta (\sqrt{-g}) \mathcal{L}_M + \sqrt{-g} \delta \mathcal{L}_M \right] \\ &\stackrel{(1)}{=} \int d^4x \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \delta g_{\mu\nu} \mathcal{L}_M + \delta \mathcal{L}_M \right] \end{aligned}$$

where in (1) we have used the property $\delta \sqrt{-g} = \frac{1}{2} \sqrt{-g} g^{\mu\nu} \delta g_{\mu\nu}$. This can be demonstrated using the following relation for an invertible square matrix g :

$$\begin{aligned} \det(g) &= e^{\text{tr}(\ln(g))} \\ \implies \delta(\det(g)) &= \delta(e^{\text{tr}(\ln g)}) = \det(g) \text{tr}(g^{-1} \delta g) \\ \implies \delta \sqrt{-g} &= \frac{1}{2} \frac{1}{\sqrt{-g}} \delta(g) = \frac{1}{2} \sqrt{-g} g^{\mu\nu} \delta g_{\mu\nu} \end{aligned}$$

From here, we equate the integrands. Recalling that $\mathcal{L}_M = -\frac{1}{4}F_{\alpha\beta}F^{\alpha\beta}$ (with $J^\mu = 0$):

$$\begin{aligned}\frac{1}{2}T^{\mu\nu}\delta g_{\mu\nu} &= \frac{1}{2}g^{\mu\nu}\delta g_{\mu\nu}\mathcal{L}_M + \delta\mathcal{L}_M = \frac{1}{2}g^{\mu\nu}\delta g_{\mu\nu}\left(-\frac{1}{4}F_{\alpha\beta}F^{\alpha\beta}\right) - \frac{1}{4}\delta\left(F_{\alpha\beta}F^{\alpha\beta}\right) \\ &= -\frac{1}{2}\frac{1}{4}\left[F_{\alpha\beta}F^{\alpha\beta}g^{\mu\nu}\delta g_{\mu\nu} + 2F_{\alpha\beta}\delta F^{\alpha\beta}\right]\end{aligned}$$

since the variation is performed with respect to the metric, meaning $\delta F_{\alpha\beta} = 0$. At this point, we calculate the term $\delta F^{\alpha\beta}$ separately:

$$\begin{aligned}\delta F^{\alpha\beta} &= \delta\left(g^{\alpha\sigma}g^{\beta\lambda}F_{\sigma\lambda}\right) = \delta(g^{\alpha\sigma})g^{\beta\lambda}F_{\sigma\lambda} + g^{\alpha\sigma}\delta(g^{\beta\lambda})F_{\sigma\lambda} \\ &= -g^{\alpha\zeta}g^{\iota\sigma}\delta g_{\zeta\iota}g^{\beta\lambda}F_{\sigma\lambda} - g^{\alpha\sigma}g^{\beta\zeta}g^{\iota\lambda}\delta g_{\zeta\iota}F_{\sigma\lambda} \\ &= -F_{\sigma\lambda}\left(g^{\iota\sigma}g^{\beta\lambda}g^{\alpha\zeta} + g^{\alpha\sigma}g^{\iota\lambda}g^{\beta\zeta}\right)\delta g_{\zeta\iota}\end{aligned}$$

Contracting this with $F_{\alpha\beta}$:

$$\begin{aligned}F_{\alpha\beta}\delta F^{\alpha\beta} &= -\left(F_{\alpha\beta}F_{\sigma\lambda}g^{\iota\sigma}g^{\beta\lambda}g^{\alpha\zeta} + F_{\alpha\beta}F_{\sigma\lambda}g^{\alpha\sigma}g^{\iota\lambda}g^{\beta\zeta}\right)\delta g_{\zeta\iota} \\ &\stackrel{(2)}{=} -\left(F_{\alpha\beta}F_{\sigma\lambda}g^{\iota\sigma}g^{\beta\lambda}g^{\alpha\zeta} + F_{\sigma\lambda}F_{\alpha\beta}g^{\sigma\alpha}g^{\iota\beta}g^{\lambda\zeta}\right)\delta g_{\zeta\iota} \\ &\stackrel{(3)}{=} -\left(-g^{\zeta\alpha}g^{\lambda\beta}F_{\alpha\beta}g^{\iota\sigma}F_{\lambda\sigma} + g^{\alpha\sigma}g^{\zeta\lambda}F_{\sigma\lambda}g^{\iota\beta}F_{\alpha\beta}\right)\delta g_{\zeta\iota} \\ &= -\left(-F^{\zeta\lambda}F^\iota_\lambda - F^{\zeta\alpha}F^\iota_\alpha\right)\delta g_{\zeta\iota} \stackrel{(4)}{=} 2F^{\zeta\lambda}F^\iota_\lambda\delta g_{\zeta\iota} = 2F^{\mu\lambda}F^\nu_\lambda\delta g_{\mu\nu}\end{aligned}$$

where in (2) we performed the index swap $\alpha \leftrightarrow \sigma$ and $\beta \leftrightarrow \lambda$ in the second term; in (3) we utilized the antisymmetry of $F_{\alpha\beta}$ and the symmetry of $g^{\alpha\beta}$; and in (4) we again used the antisymmetry while swapping $\alpha \leftrightarrow \lambda$. Substituting this result back into the action variation:

$$\begin{aligned}\frac{1}{2}T^{\mu\nu}\delta g_{\mu\nu} &= -\frac{1}{8}F_{\alpha\beta}F^{\alpha\beta}g^{\mu\nu}\delta g_{\mu\nu} - \frac{1}{4}\left(2F^{\mu\lambda}F^\nu_\lambda\delta g_{\mu\nu}\right) \\ &= \frac{1}{2}\left[-F^{\mu\lambda}F^\nu_\lambda - \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}\right]\delta g_{\mu\nu}\end{aligned}$$

In Heaviside-Lorentz units, lowering the indices of the first term results in the symmetric form:

$$T^{\mu\nu} = F^\mu_\lambda F^{\nu\lambda} - \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}$$

2. Scalar Field Energy-Momentum Tensor

Next, we calculate the energy-momentum tensor for a scalar field defined by the action:

$$S[\phi] = \int d^4x \sqrt{-g(x)} \left[-\frac{1}{2}g^{\mu\nu}(x)\partial_\mu\phi(x)\partial_\nu\phi(x) - V(\phi(x)) \right].$$

Applying the same strategy, assuming $\delta\phi = 0$:

$$\delta S = \int d^4x \left(\delta(\sqrt{-g})\mathcal{L} + \sqrt{-g}\delta\mathcal{L} \right)$$

$$\begin{aligned}
&= \int d^4x \sqrt{-g} \left(\frac{1}{2} g_{\rho\lambda} \delta g^{\rho\lambda} \mathcal{L} - \frac{1}{2} \delta g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \right) \\
&= -\frac{1}{2} \int d^4x \sqrt{-g} (\partial_\mu \phi \partial_\nu \phi - g_{\mu\nu} \mathcal{L}) \delta g^{\mu\nu} \\
\Rightarrow T^{\mu\nu} &= \partial^\mu \phi \partial^\nu \phi + g^{\mu\nu} \mathcal{L} = \partial^\mu \phi \partial^\nu \phi + g^{\mu\nu} \left[-\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi - V(\phi) \right]
\end{aligned}$$

3. Physical Limit and Energy Density

If $\phi = \phi(t)$ depends only on time, the temporal component of the tensor becomes:

$$\begin{aligned}
T^{00} &= \partial^0 \phi \partial^0 \phi + g^{00} \left[-\frac{1}{2} g^{00} \partial_0 \phi \partial_0 \phi - V \right] \\
&= \left(g^{00} \partial_0 \phi \right)^2 - \frac{1}{2} \left(g^{00} \partial_0 \phi \right)^2 - g^{00} V = \frac{1}{2} \left(g^{00} \dot{\phi} \right)^2 - g^{00} V
\end{aligned}$$

In the Minkowski limit ($g^{00} = -1$), this yields the physical energy density:

Scalar Field Energy Density

The energy-momentum tensor reduces to the sum of the kinetic and potential energy:

$$T^{00} = \frac{1}{2} \dot{\phi}^2 + V(\phi) = \rho_\phi$$